

Request for Expert Content Review

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Registered Medical Technologist

Dear Ms. Montaña:

Greetings!

We are fourth-year Bachelor of Science in Computer Science students from West Visayas State University. As part of our undergraduate thesis, we developed **URITECT**, an offline smartphone-based urinalysis dipstick screening and clinical decision-support application.

We respectfully request your professional review of the proposed urinalysis dipstick mappings and literature-derived likelihood-ratio parameters used in URITECT's sex-specific Bayesian urinary tract infection (UTI) research component.

URITECT serves eligible male and female adult users for ten-analyte strip scanning, safety assessment, and renal follow-up. Its UTI research component uses separate evidence pathways because the available studies differ by sex:

- The female candidate model uses evidence derived principally from symptomatic, nonpregnant women.
- The male candidate model uses culture-referenced evidence from symptomatic men and is limited to nitrite and leukocyte esterase. Female blood and symptom likelihood ratios are not applied to men.

A likelihood ratio describes how a finding changes the evidence supporting UTI. A value greater than 1 increases the evidence, a value below 1 decreases it, and a value of 1 produces no update. An LR is not itself a percentage.

The attached form shows every candidate value, its source, the exact strip mapping, and how URITECT prevents correlated findings from being counted more than once. The resulting estimates are provisional, literature-derived, and not locally calibrated. They do not diagnose UTI, recommend treatment, replace urine culture, or replace professional judgment. URITECT does not convert the estimates into Low, Moderate, or High categories.

We ask you to assess whether the laboratory mappings, candidate parameters, handling of missing or unreliable results, and screening workflow are relevant and acceptable for the stated research purpose. Further physician, clinical, statistical, and culture-confirmed patient-level validation is required before the estimates can be described as clinically validated probabilities.

Thank you for sharing your professional time and expertise.

Respectfully,

The URITECT Research Team

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